

Experiment - 10

Data Visualization III

Download the Iris flower dataset or any other dataset into a DataFrame. (e.g., <https://archive.ics.uci.edu/ml/datasets/Iris>). Scan the dataset and give the inference as:

1. List down the features and their types (e.g., numeric, nominal) available in the dataset.
2. Create a histogram for each feature in the dataset to illustrate the feature distributions.
3. Create a box plot for each feature in the dataset.
4. Compare distributions and identify outliers.

```
In [1]: import pandas as pd
import seaborn as sns
```

```
In [2]: data=pd.read_csv(r"D:\College\TE\SEM-2\Practical\DSBDA\10\Iris.csv")
```

```
In [3]: data.head()
```

Out[3]:

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	1	5.1	3.5	1.4	0.2	Iris-setosa
1	2	4.9	3.0	1.4	0.2	Iris-setosa
2	3	4.7	3.2	1.3	0.2	Iris-setosa
3	4	4.6	3.1	1.5	0.2	Iris-setosa
4	5	5.0	3.6	1.4	0.2	Iris-setosa

```
In [4]: data.describe()
```

Out[4]:

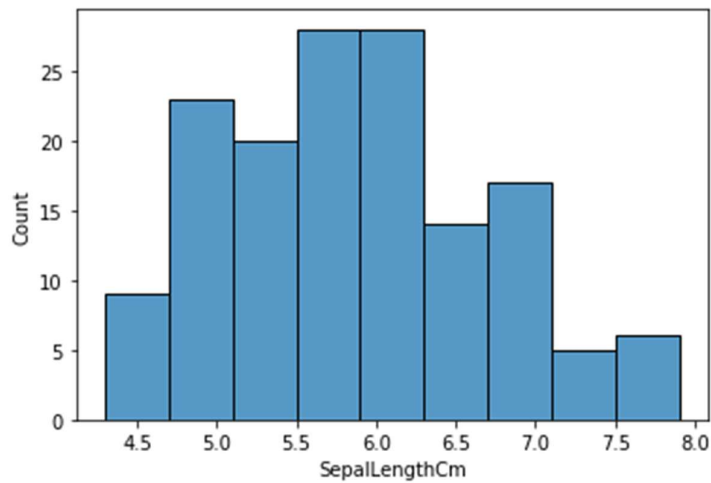
	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm
count	150.000000	150.000000	150.000000	150.000000	150.000000
mean	75.500000	5.843333	3.054000	3.758667	1.198667
std	43.445368	0.828066	0.433594	1.764420	0.763161
min	1.000000	4.300000	2.000000	1.000000	0.100000
25%	38.250000	5.100000	2.800000	1.600000	0.300000
50%	75.500000	5.800000	3.000000	4.350000	1.300000
75%	112.750000	6.400000	3.300000	5.100000	1.800000
max	150.000000	7.900000	4.400000	6.900000	2.500000

```
In [5]: data.info()
```

```
<class 'pandas.core.frame.DataFrame'>  
RangeIndex: 150 entries, 0 to 149  
Data columns (total 6 columns):  
Id                150 non-null int64  
SepalLengthCm    150 non-null float64  
SepalWidthCm     150 non-null float64  
PetalLengthCm    150 non-null float64  
PetalWidthCm     150 non-null float64  
Species          150 non-null object  
dtypes: float64(4), int64(1), object(1)  
memory usage: 7.2+ KB
```

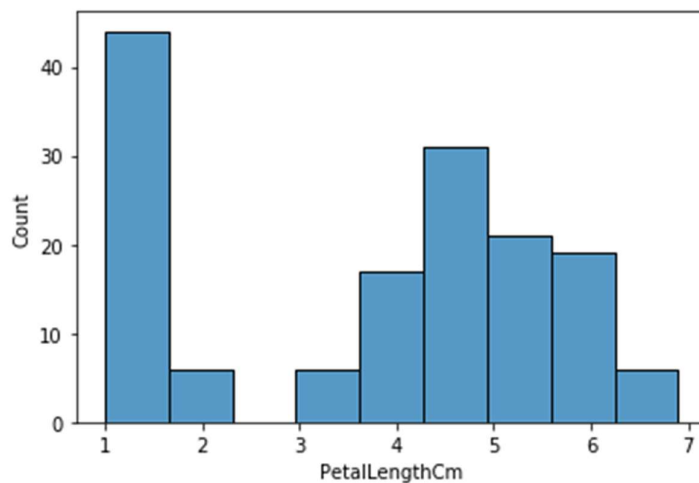
```
In [6]: sns.histplot(data=data, x="SepalLengthCm")
```

```
Out[6]: <matplotlib.axes._subplots.AxesSubplot at 0x2274e737808>
```



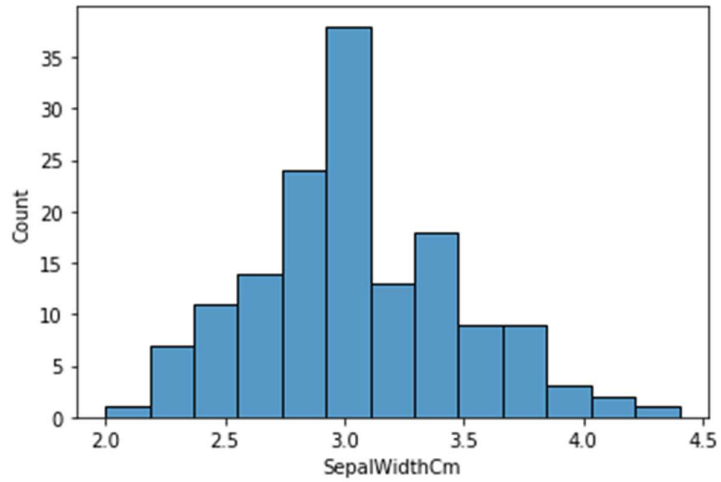
```
In [7]: sns.histplot(data=data, x="PetalLengthCm")
```

```
Out[7]: <matplotlib.axes._subplots.AxesSubplot at 0x2274eab8888>
```



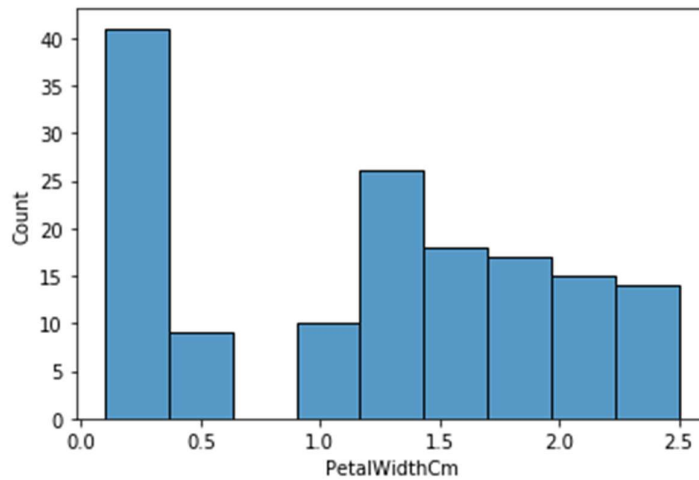
```
In [8]: sns.histplot(data=data, x="SepalWidthCm")
```

```
Out[8]: <matplotlib.axes._subplots.AxesSubplot at 0x2274fb0c148>
```



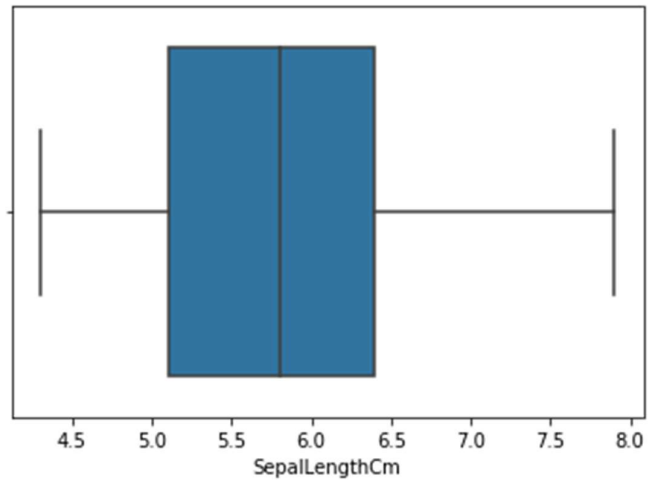
```
In [9]: sns.histplot(data=data, x="PetalWidthCm")
```

```
Out[9]: <matplotlib.axes._subplots.AxesSubplot at 0x2274fbb9088>
```



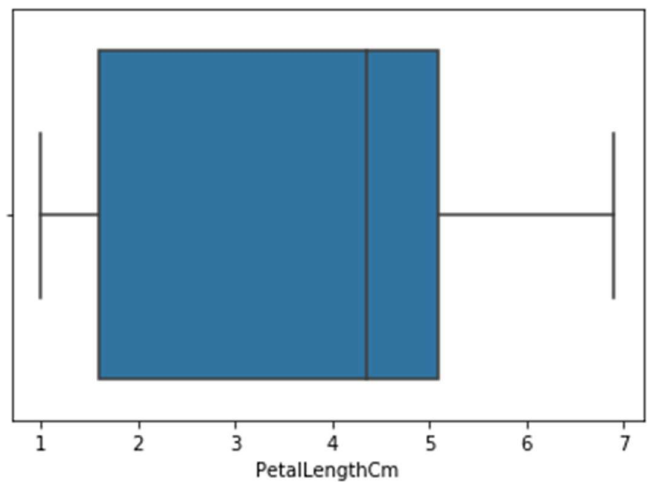
```
In [10]: sns.boxplot(x="SepalLengthCm", data=data)
```

```
Out[10]: <matplotlib.axes._subplots.AxesSubplot at 0x2274fc47d88>
```



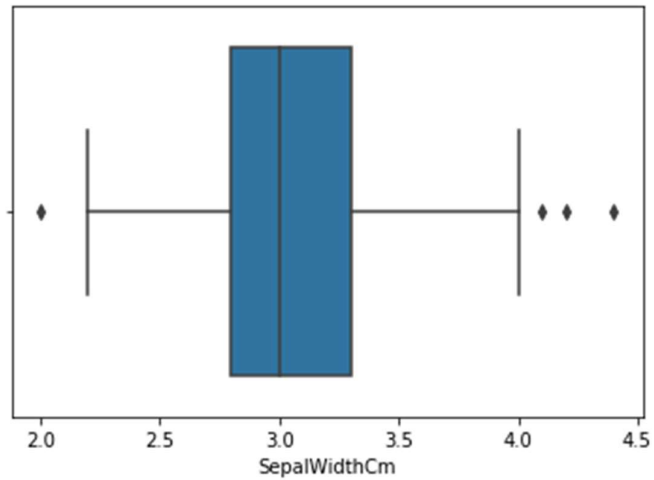
```
In [11]: sns.boxplot(x="PetalLengthCm", data=data)
```

```
Out[11]: <matplotlib.axes._subplots.AxesSubplot at 0x2274fc977c8>
```



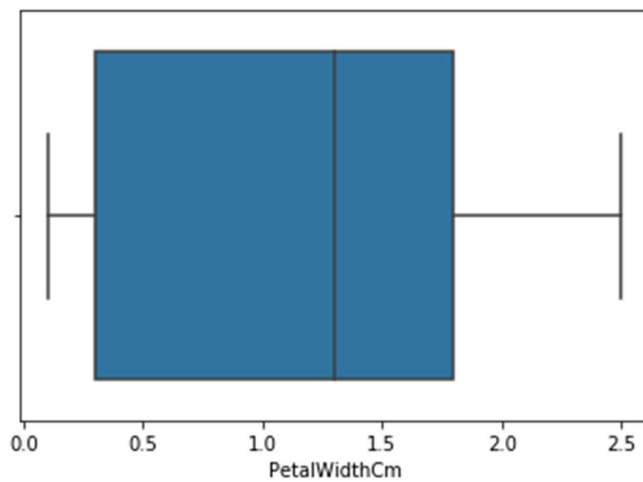
```
In [12]: sns.boxplot(x="SepalWidthCm", data=data)
```

```
Out[12]: <matplotlib.axes._subplots.AxesSubplot at 0x2274fd2a3c8>
```



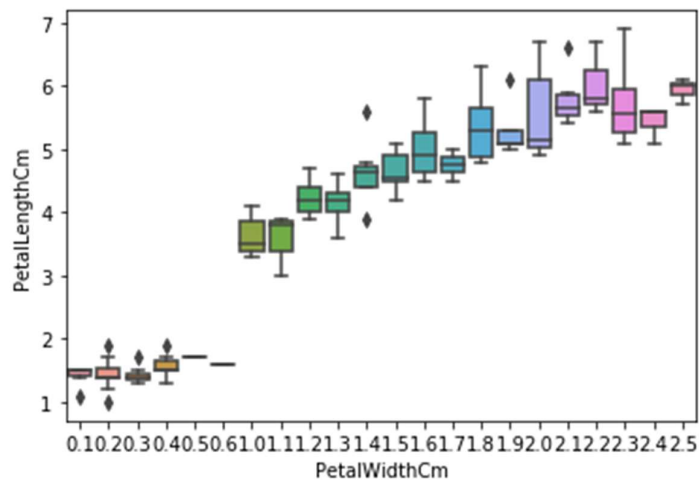
```
In [13]: sns.boxplot(x="PetalWidthCm", data=data)
```

```
Out[13]: <matplotlib.axes._subplots.AxesSubplot at 0x2274fc58548>
```



```
In [14]: sns.boxplot(x="PetalWidthCm", y="PetalLengthCm", data=data)
```

```
Out[14]: <matplotlib.axes._subplots.AxesSubplot at 0x2274fdd4648>
```



```
In [15]: sns.boxplot(x="SepalWidthCm", y="SepalLengthCm", data=data)
```

```
Out[15]: <matplotlib.axes._subplots.AxesSubplot at 0x2275004c188>
```

